Practical-9

AIM: Implementation of SUBNETTING in CISCO PACKET TRACER Simulator

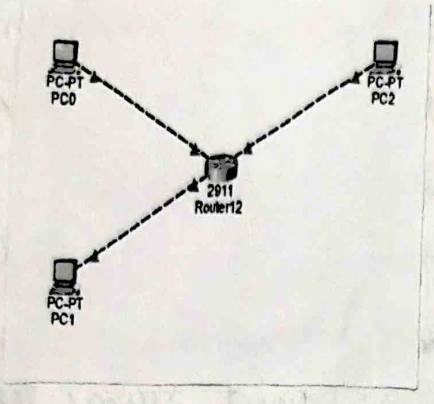
Classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet masks that are not just the default masks for each IP class. This means that we can divide our IP address space into smaller subnets, which can be useful when we have a limit number of IP address but need to create multiple networks.

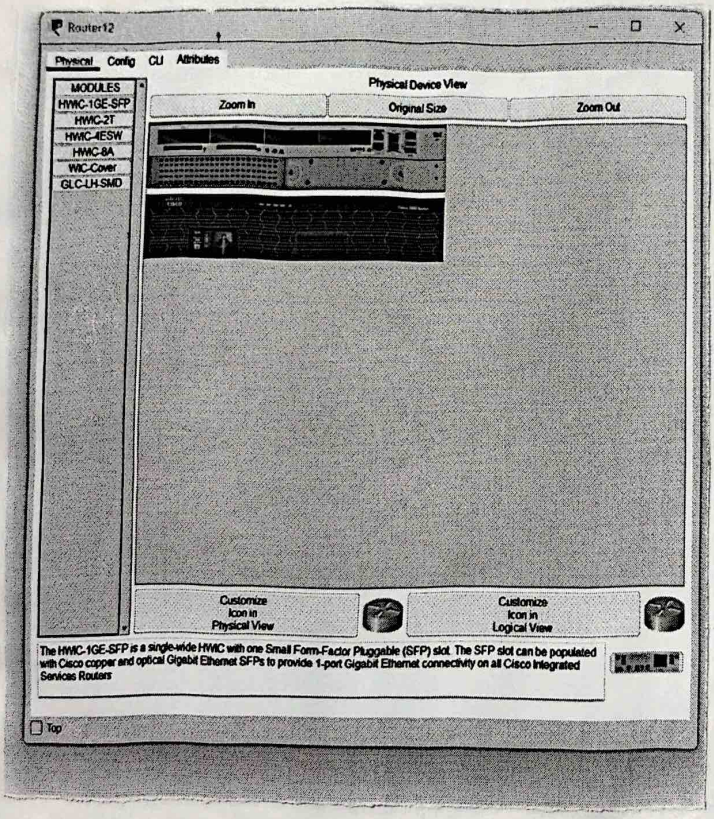
SUBNETTING:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 address for end devices, the switch, and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host address each.

The IP addressing for the network shown in the topology can be as follows:

- Router R1:
- Gigabit Ethernet 0/0: 192.168.1.1
- Gigabit Ethernet 0/1: 192.168.2.1
- Switch S1:
- Fast Ethernet 0/1: 192.168.1.0/27


PC-PT
PC0
PC-PT
PC2
2911
Router12
PC-PT
PC1


Router12
Physical
Config
CLI
Attributes
Physical Device View
MODULES
HWIC-1GE-SFP
HWIC-2T
HWIC-4ESW
HWIC-8A
WIC-Cover
GLC-LH-SMD
Zoom In
Original Size
Zoom Out
Customize
Icon in
Physical View
Customize
Icon in
Logical View
The HWIC-1GE-SFP is a single-wide HWIC with one Small Form-Factor Pluggable (SFP) slot. The SFP slot can be populated with Cisco copper and optical Gigabit Ethernet SFPs to provide 1-port Gigabit Ethernet connectivity on all Cisco Integrated Services Routers
Top

- Pc1 : 192.168.1.11
- Pc2 : 192.168.1.12
- Pc3 : 192.168.1.13
- Pc4 : 192.168.1.14
- Pc5 : 192.168.1.15
- Fast Ethernet 0/2 ; 192.168.2.0/27
- Pc1 ; 192.168.2.11
- Pc2 ; 192.168.2.12
- Pc3 . 192.168.2.13
- Pc4 ; 192.168.2.14
- Pc5 : 192.168.2.15
- Router R2 ;
- Fast Ethernet 0/0 : 192.168.3.1
- Fast Ethernet 0/1 ; 192.168.4.1
- Switch S2 ;
- Fast Ethernet 0/1 ; 192.168.3.0/27
- Pc1 : 192.168.3.11
- Pc2 ; 192.168.3.12
- Pc3 ; 192.168.3.13
- Pc4 ; 192.168.3.14
- Pc5 ; 192.168.3.15
- Fast Ethernet 0/2 ; 192.168.4.0/27
- Pc1 ; 192.168.4.11
- Pc2 ; 192.168.4.12
- Pc3 ; 192.168.4.13
- Pc4 ; 192.168.4.14
- Pc5 ; 192.168.4.15

```
Router(config)#int g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
```

```
Router(config)#int g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
```

```
Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shitdown
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     192.168.3.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router>running-config
Translating "running-config"...domain server (255.255.255.255) % Name lookup aborted
Router>en
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#show running-conf
Building configuration...

Current configuration : 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
ip cef
no ipv6 cef
!
!
!
license udi pid CISCO2911/K9 sn FTX1524WFBP-
!
!
!
spanning-tree mode pvst
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
```

CONFIGURING THE DEVICES:

Start configuring them. Right-click on the router and select "CLI".

the command-line interface (CLI) for the router. In the CLI, enter the following commands:

```
# enable
# configure terminal
# interface FastEthernet 0/0
# ip address {IP address} {subnet mask}
# no shutdown
# exit
  interface FastEthernet 0/1
  ip address {IP address} {Subnet mask}
  no shutdown
  exit
```

Next, we will configure the switch, Right-click on the switch and select "CLI". In the CLI, enter the following commands:

```
enable
configure terminal
interface FastEthernet 0/1
SwitchPort mode access
exit
Interface FastEthernet 0/2
```

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

Automatic Static

IPv6 Address /

Link Local Address FE80::260:3EFF:FEC3:2B4A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication MD5

Username

Password

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ip config
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::260:3EFF:FEC3:2B4A
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.1.2
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                         192.168.2.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0
```

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#show running-conf
Building configuration...

Current configuration : 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

SwitchPort mode access
exit

These Commads configure the Switch to operate in access mode on its two Ports, which are connected to the two PCs.

To Configure the Gigabit Ethernet interface on the router, you can follow these steps:

1. Right-click on the router and Select "CLI".
2. Enter the following commands:

Configure terminal
interface GigabitEthernet 0/0
IP address {IP address} {Subnet mask} no Shutdown
exit.

TESTING THE NETWORK:

Open a Command Prompt on each PC and try to Ping the other PC. If the Ping is Successful, then the network is functioning Properly.

Result :-

Thus the Subnet implementation is performed in CISCO packet tracer simulator.

```
Router#show startup-conf
Using 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524WFBP-
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!

Router#
```

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=12
Reply from 192.168.2.2: bytes=32 time<1ms TTL=12
Reply from 192.168.2.2: bytes=32 time<1ms TTL=12

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC2

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- ○ DHCP
- ● Static

IPv4 Address	192.168.3.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	0.0.0.0